

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 1

Notation 1. We will use *summation notation*: The symbol

$$\sum_{k=1}^n a_k$$

has to be read like a for loop: The entries $a_1, a_2, \dots, a_n$ are numbers that are indexed by their subscripts, and the symbol above is short for the iterated sum

$$a_1 + a_2 + \dots + a_n.$$

(1) Suppose that we have $a, b \in \mathbb{Z}$.

(a) If a, b each leave a remainder of 1 after dividing by 4, what can you say about the remainder that ab leaves when divided by 4?

Solution. Our assumption translates concretely to saying that there exist integers $k, l \in \mathbb{Z}$ with $a = 4k + 1$ and $b = 4l + 1$. The product is

$$ab = (4k + 1)(4l + 1) = 16kl + 4l + 4k + 1 = 4(4kl + l + k) + 1,$$

which clearly has remainder 1 when divided by 4.

(b) If a, b each leave a remainder of 3 after dividing by 4, what can you say about the remainder that ab leaves when divided by 4?

Solution. Just as before, except that we have $a = 4k + 3$ and $b = 4l + 3$ with the product given by

$$ab = (4k + 3)(4l + 3) = 16kl + 12k + 12l + 9 = 4(4kl + 3k + 3l + 2) + 1,$$

which has remainder 1 when divided by 4.

Remark 1. In the notation introduced in Lecture 4, we can encapsulate this as follows:

- (a) If $a \equiv 1 \pmod{4}$ and $b \equiv 1 \pmod{4}$, then $ab \equiv 1 \pmod{4}$.
- (b) If $a \equiv 3 \pmod{4}$ and $b \equiv 3 \pmod{4}$, then $ab \equiv 1 \pmod{4}$.

These are of course reasonable things to be true!

(2) Use the previous problem to show that there are infinitely many primes (such as 3, 7, 11, 19, ...) that leave a remainder of 3 after dividing by 4.

Hint: This uses a variation of the construction used in Euclid's proof of the infinitude of all primes: If $p_1, \dots, p_r$ is a finite set of primes, consider the prime factorization of $N = 4p_1 \cdots p_r - 1$.

Solution. Let X be the set of primes that leave a remainder of 3 after dividing by 4. We will show that, for any finite subset $\{p_1, \dots, p_r\} \subset X$, we can find $q \in X$ such that q does not belong to $\{p_1, \dots, p_r\}$. This is the concrete manifestation of the infinitude of X .

To do this, as the hint says, we will consider the number

$$N = 4p_1 \cdots p_r - 1.$$

As we argued in Lecture 2, this natural number is a product of prime numbers: $N = q_1 \cdots q_m$. Furthermore, we know the following things:

- N is odd (why?) meaning none of the primes q_j can be 2;
- None of the primes $q_1, \dots, q_m$ belongs to the finite set $\{p_1, \dots, p_r\}$: this is because, *by its construction*, none of the primes in this finite set can divide N (why?);
- N leaves a remainder of 3 when divided by 4. In the notation from Lecture 4, we have

$$N \equiv 3 \pmod{4}.$$

Now, the first point means that each q_i is congruent to either 1 or 3 mod 4 (why?). If *all* of them happened to be 1 mod 4, then problem 1 shows that their product (which is N) would also be 1 mod 4. But the last point tells us that N is congruent to 3 mod 4. The only way for this to be possible is if at least *one* of $q_1, \dots, q_m$ is congruent to 3 mod 4. Up to reindexing the subscripts, we can assume that $q_1 \equiv 3 \pmod{4}$. Now, the second bullet point tells us that $q = q_1$ is a *new* element of X that's not one of the already known $p_1, \dots, p_r$.

This completes the proof.

(3) Prove the following formulas using induction:

(a)

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6};$$

Solution. **Base case:** When $n = 1$, the left hand side is $1^2 = 1$ and the right hand side is $\frac{1(1+1)(2+1)}{6} = \frac{2 \cdot 3}{6} = 1$. These are clearly equal.

Inductive hypothesis: We have an equality

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

for n .

Inductive step: We need to verify the equality with n replaced by $n+1$. For this we add $(n+1)^2$ to both sides of our inductive hypothesis. This gives us

$$\sum_{k=1}^{n+1} k^2$$

on the left hand side, and

$$\frac{n(n+1)(2n+1)}{6} + (n+1)^2$$

on the right. Let us manipulate the RHS:

$$\begin{aligned} \frac{n(n+1)(2n+1)}{6} + (n+1)^2 &= \frac{n(n+1)(2n+1) + 6(n+1)^2}{6} \\ &= \frac{(n+1)(n(2n+1) + 6(n+1))}{6} \\ &= \frac{(n+1)(2n^2 + n + 6n + 6)}{6} \\ &= \frac{(n+1)(2n^2 + 7n + 6)}{6}. \end{aligned}$$

We would like for this to be equal to:

$$\begin{aligned} \frac{(n+1)(n+2)(2(n+1)+1)}{6} &= \frac{(n+1)(n+2)(2n+2+1)}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6} \\ &= \frac{(n+1)(2n^2 + 7n + 6)}{6}. \end{aligned}$$

The two are now clearly the same thing!

(b)

$$\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}.$$

Solution. Same as the previous problem. The inductive step comes down to the algebraic manipulations

$$\begin{aligned} \frac{n^2(n+1)^2}{4} + (n+1)^3 &= \frac{n^2(n+1)^2 + 4(n+1)^3}{4} \\ &= \frac{(n+1)^2(n^2 + 4(n+1))}{4} \\ &= \frac{(n+1)^2(n^2 + 4n + 4)}{4} \\ &= \frac{(n+1)^2(n+2)^2}{4}. \end{aligned}$$

- (4) Prove that $2^{2n+1} + 1$ is a multiple of 3 for every $n \in \mathbb{N}$.

Solution. **Base case:** When $n = 1$, this is $2^3 + 1 = 9$, which is $3 \cdot 3$.

Inductive hypothesis: $2^{2n+1} + 1 = 3k$ for some $k \in \mathbb{N}$.

Inductive step: Our inductive hypothesis can be rewritten as

$$2^{2n+1} = 3k - 1.$$

Now, if we plug in $n + 1$ for n , we get

$$2^{2(n+1)+1} + 1 = 2^{2n+3} + 1 = 4 \cdot 2^{2n+1} + 1 = 4(3k - 1) + 1 = 12k - 4 + 1 = 12k - 3,$$

and this is of course a multiple of 3.

Remark 2. We can also prove this directly using mod-3 arithmetic. We have

$$2 \equiv -1 \pmod{3} \Rightarrow 2^{2n+1} \equiv (-1)^{2n+1} \equiv -1 \pmod{3}.$$

So adding 1 to this gives us $0 \pmod{3}$, and so shows that $2^{2n+1} + 1$ is a multiple of 3.

- (5) Prove that for every $n \in \mathbb{N}$, we have

$$\sum_{k=1}^n \frac{1}{k^2} \leq 2 - \frac{1}{n}.$$

Solution. **Base case:** When $n = 1$, the LHS is just $1/1^2 = 1$, while the RHS is $2 - \frac{1}{1} = 1$, and we clearly have $1 \leq 1$.

Inductive hypothesis:

$$\sum_{k=1}^n \frac{1}{k^2} \leq 2 - \frac{1}{n}.$$

Inductive step: Add $\frac{1}{(n+1)^2}$ to both sides to get the inequality

$$\sum_{k=1}^{n+1} \frac{1}{k^2} \leq 2 - \frac{1}{n} + \frac{1}{(n+1)^2}.$$

To finish, we must show that

$$2 - \frac{1}{n} + \frac{1}{(n+1)^2} \leq 2 - \frac{1}{n+1}.$$

This would follow if we knew:

$$\frac{1}{(n+1)^2} \leq \frac{1}{n} - \frac{1}{n+1}.$$

The RHS here is

$$\frac{1}{n} - \frac{1}{n+1} = \frac{n+1-n}{n(n+1)} = \frac{1}{n(n+1)}$$

Now, the desired inequality

$$\frac{1}{(n+1)^2} \leq \frac{1}{n(n+1)}$$

is true because the denominator on the LHS is clearly larger than the one on the RHS.

(6) Use l'Hôpital's rule and induction to prove that

$$\lim_{n \rightarrow \infty} \frac{x^n}{e^x} = 0$$

for all $n \in \mathbb{N}$. You may assume that the equality is true when $n = 0$.

Solution. Base case: Though the problem asks you to show this for all $n \in \mathbb{N}$, there is actually only additional generality in showing it for all *non-negative* integers, including 0. So we will start the base case at $n = 0$, where we are told we can assume the truth of the equality.

Inductive hypothesis:

$$\lim_{n \rightarrow \infty} \frac{x^n}{e^x} = 0$$

Inductive step:

Note that x^{n+1} and e^x are going to ∞ as $x \rightarrow \infty$. Therefore, l'Hôpital's rule applies and tells us that we have

$$\lim_{n \rightarrow \infty} \frac{x^{n+1}}{e^x} = \lim_{n \rightarrow \infty} \frac{(n+1)x^n}{e^x} = (n+1) \lim_{n \rightarrow \infty} \frac{x^n}{e^x} = (n+1) \cdot 0 = 0.$$

In the penultimate equality, we have used our inductive hypothesis.